Methodology Alignment Analysis Template

IMMU-201 | Phase II Immunotherapy Trial

Synthetic training data - educational use only.

Instructions

Use the IMMU-201 Protocol Summary Document as your central source. Map the study objective to the endpoint and data collection approach, then assess whether those elements align. Support each conclusion with study-specific evidence rather than general statements.

- Review the model feedback entry first. It uses a different synthetic protocol and is provided only to demonstrate expected depth and structure.
- Complete every section for IMMU-201 using information from the Protocol Summary.
- Identify the relevant objective, endpoint, timing, and data collection method.
- Explain why the methodology does or does not adequately answer the stated objective.
- When you identify a gap, describe specifically what is missing or misaligned.

Model Feedback Entry - Different Protocol

Model protocol: MIGRAINE-102, a fictional Phase II migraine-prevention study. This example is not part of IMMU-201.

Study Objective	Evaluate whether Compound X reduces monthly migraine days after 12 weeks of treatment compared with placebo.
Endpoint	Change in mean monthly migraine days from baseline to Week 12.
Data Collection	Participants record headache occurrence in a daily electronic diary; monthly migraine-day totals are derived from diary entries at baseline and during treatment.
Alignment Assessment	Strong alignment.
Evidence / Rationale	The objective asks whether the intervention reduces migraine frequency, and the endpoint directly measures change in migraine days over the same 12-week treatment period. Daily diary collection provides repeated participant-level data needed to calculate the endpoint.
Feedback	The objective, endpoint, and collection method

	are directly connected. The analysis would be stronger if the protocol also specified the minimum diary-completion threshold required for a participant's monthly migraine-day data to be considered evaluable.

What to notice in the model: The feedback does more than label the methodology as aligned. It identifies the exact objective, endpoint, timing, and collection method, explains the connection among them, and gives a specific improvement point.

Section 1: Primary Objective-to-Endpoint Assessment

Primary Objective	To evaluate the efficacy of IMT-201 compared with placebo in reducing disease activity from baseline to Week 24.
Corresponding Primary Endpoint	Change from baseline to Week 24 in the protocol-defined rheumatoid arthritis disease activity score.
Endpoint Timing	Baseline/Day 1 (before randomization and first injection) and Week 24, which is 4 weeks after the final injection at Week 20. Disease assessments are also performed at the Week 4–20 visits.
Data Collection Method(s)	Clinical disease activity assessments (disease activity examinations, including joint examination) performed by blinded, delegated study staff at Baseline/Day 1, Weeks 4–20, and the Week 24 primary endpoint visit. Concomitant medication review at each visit documents that background therapy remains stable.
Alignment Rating	Strong Alignment
Evidence and Rationale	The objective asks whether IMT-201 reduces disease activity compared with placebo from baseline to Week 24. The primary endpoint measures exactly that construct (the rheumatoid arthritis disease activity score), as a change, over exactly that interval (baseline to Week 24). The design supplies the comparison the objective requires: approximately 120 participants randomized 1:1 to IMT-201 or matched placebo, with participants, investigators, and coordinators involved in

	outcome assessment blinded (Section 4). The Visit Overview (Section 11) schedules a disease activity assessment at Baseline/Day 1 and a "primary endpoint disease assessment" at Week 24, so both values needed to calculate the endpoint are collected.
Written Feedback	The primary objective, primary endpoint, timing, and collection method are directly connected: the same outcome and the same timeframe appear in all three, and the randomized, double-blind, placebo-controlled design allows a between-group difference in that endpoint to be attributed to IMT-201. Blinding is especially important here because the disease activity examination relies on assessor judgment. The alignment would be easier to verify if the summary named the instrument behind the "protocol-defined rheumatoid arthritis disease activity score" and its components, and stated who performs the joint assessment. The site should confirm these in the full protocol, along with the permitted window for the Week 24 visit, because a missed or out-of-window Week 24 assessment directly affects the primary endpoint.

Section 2: Secondary Objective-to-Endpoint Assessment

Select at least two secondary objectives from the Protocol Summary and map each one to the endpoint and data collection approach used to evaluate it.

Secondary Objective 1

Secondary Objective	To evaluate change in participant physical function from baseline to Week 24.
Corresponding Endpoint	Change from baseline to Week 24 in a validated participant physical-function questionnaire score.
Data Collection Method(s)	Participant-reported outcome: participants complete the physical-function questionnaire at Baseline/Day 1, at Weeks 4–20 "as scheduled," and at Week 24 (Sections 7, 8, and 11).
Alignment Rating	Strong Alignment

Evidence and Rationale	The objective and the endpoint name the same construct (physical function), the same measure of effect (change), and the same interval (baseline to Week 24). Section 8 confirms that participant questionnaires are the procedure that "measures physical function and pain," and Section 11 lists questionnaires at Baseline/Day 1 and "physical-function and pain assessments" at Week 24. A validated questionnaire is an appropriate way to measure function because function is experienced by the participant, and double-blinding protects a self-reported measure from expectation bias.
Written Feedback	The physical-function objective is fully answered by its endpoint: a validated instrument, completed by the participant, at the two timepoints the objective specifies. The summary does not name the questionnaire or state which interim visits include it ("as scheduled"). The site should confirm the instrument, version, and visit-by-visit schedule in the full protocol, and plan for questionnaires to be completed privately and before the participant discusses clinical findings with staff.

Secondary Objective 2

Secondary Objective	To evaluate change in participant-reported pain from baseline to Week 24.
Corresponding Endpoint	Change from baseline to Week 24 in participant-reported pain using a 0–10 numeric rating scale.
Data Collection Method(s)	Participant-reported outcome: participants rate pain on a 0–10 numeric rating scale at Baseline/Day 1, at Weeks 4–20 "as scheduled," and at Week 24 (Sections 7, 8, and 11).
Alignment Rating	Strong Alignment
Evidence and Rationale	The objective asks about change in participant-reported pain from baseline to Week 24. The endpoint measures change in participant-reported pain over the same interval and specifies the tool, a 0–10 numeric rating scale.

	The source of the data (the participant) is the one the objective names. Section 11 places questionnaires at Baseline/Day 1 and pain assessment at Week 24, so both values needed for the change score are collected under blinded conditions.
Written Feedback	The pain objective and its endpoint match on construct, reporter, measure, and timing, and this is the most precisely specified of the secondary endpoints because the scale is named. One detail to confirm in the full protocol is the recall period for the rating (for example, current pain versus pain over the past week). Coordinators need to apply the same instruction at every visit for the baseline and Week 24 scores to be comparable.

Secondary Objective 3

Secondary Objective	To characterize the safety and tolerability of IMT-201, including adverse events, serious adverse events, injection-site reactions, clinically significant laboratory abnormalities, and treatment discontinuations.
Corresponding Endpoint	Incidence and severity of treatment-emergent adverse events and serious adverse events through Week 24.
Data Collection Method(s)	Adverse-event assessment, vital signs, safety laboratory testing, and concomitant medication review at protocol-specified visits; AE/SAE assessment and labs at every Week 4–20 visit and a safety evaluation at Week 24 (Sections 7 and 11).
Alignment Rating	Partial Alignment
Evidence and Rationale	The objective lists five safety components. The endpoint explicitly covers two of them, adverse events and serious adverse events, and its timeframe (through Week 24) matches the treatment period. The data collection approach is broader than the endpoint: safety laboratory testing and vital

	signs are collected at protocol-specified visits, so the data needed to identify clinically significant laboratory abnormalities exist. However, the summary lists no endpoint for laboratory abnormalities, injection-site reactions, or treatment discontinuations.
Written Feedback	Gap: the safety and tolerability objective names injection-site reactions, clinically significant laboratory abnormalities, and treatment discontinuations, but the safety endpoint ("incidence and severity of treatment-emergent adverse events and serious adverse events through Week 24") does not name them. Injection-site reactions and laboratory abnormalities may be captured as adverse events, but the summary does not say so, and discontinuation is a tolerability measure that is not an adverse event at all. Stronger alignment would add endpoints that mirror the objective: incidence of injection-site reactions, incidence of clinically significant laboratory abnormalities, and the proportion of participants who discontinue treatment, with reasons. For the site, this means recording injection-site findings, the investigator's assessment of abnormal laboratory results, and the reason for any discontinuation in source at every visit, regardless of how the endpoint is worded.

Section 3: Methodology Elements Map

Methodology Element	Study-Specific Information	How It Supports the Objective
Design type	Phase II, multicenter, randomized, double-blind, placebo-controlled, parallel-group clinical trial (Section 4).	A controlled, prospective comparison is required to evaluate "efficacy of IMT-201 compared with placebo." Parallel groups keep each participant on one assignment for the full baseline-to-Week 24 interval.
Target population	Adults aged 18 through 75 with rheumatoid arthritis diagnosed	Matches the population in the research question. Requiring

	at least 6 months before screening, moderate-to-severe active disease at screening and baseline, and an inadequate response to background therapy held stable for at least 8 weeks before randomization (Section 5).	active disease at baseline leaves room to measure a reduction, and exclusions for active serious infection, tuberculosis, and immunodeficiency protect participants and limit confounding of the safety objective.
Intervention	IMT-201, an investigational immunotherapy, by subcutaneous injection every 4 weeks through Week 20 (six doses: Day 1 and Weeks 4, 8, 12, 16, and 20).	Provides consistent exposure across the treatment period, so the change in disease activity at Week 24 reflects a full course of treatment.
Comparator	Matched placebo by subcutaneous injection on the same schedule; both arms continue stable protocol-permitted background therapy.	Separates the effect of IMT-201 from natural disease fluctuation, the response to injections and study attention, and the effect of background therapy. Matching the appearance and schedule makes blinding possible.
Randomization	Approximately 120 eligible participants randomized 1:1 to IMT-201 or matched placebo at Baseline/Day 1, after eligibility is confirmed and documented.	Chance assignment is expected to balance known and unknown characteristics between arms, so a between-group difference in the primary endpoint can be attributed to treatment.
Blinding	Double-blind: participants, investigators, coordinators involved in outcome assessment, and designated study personnel remain blinded unless emergency unblinding is medically necessary.	Prevents knowledge of assignment from influencing the joint examination, participant-reported function and pain scores, and the reporting and assessment of adverse events.
Assessment schedule	Screening, Baseline/Day 1, and Weeks 4, 8, 12, 16, 20, and 24. Disease assessments, labs, medication review, and AE/SAE assessment at Weeks 4–20; primary endpoint	Baseline and Week 24 supply the two values needed for every change-from-baseline endpoint. Visits every 4 weeks support dosing, ongoing safety monitoring, and confirmation that background therapy

	assessment at Week 24.	remains stable.
Primary endpoint	Change from baseline to Week 24 in the protocol-defined rheumatoid arthritis disease activity score.	Directly measures the outcome (disease activity), the effect (reduction), and the timeframe (baseline to Week 24) stated in the primary objective and predicted by the hypothesis.

Section 4: Data Collection and Timing Assessment

Assessment timing: Are the endpoint assessments scheduled at a time that matches the stated objective? Explain.

Yes. The primary objective and the two participant-reported secondary objectives are all framed as change "from baseline to Week 24," and the corresponding endpoints are assessed at Baseline/Day 1 and Week 24. Section 11 confirms a disease activity assessment and questionnaires at Baseline/Day 1 and a "primary endpoint disease assessment" plus "physical-function and pain assessments" at Week 24.

The final injection is given at Week 20, so the Week 24 assessment falls one full 4-week dosing interval after the last dose and reflects the completed treatment course. Safety is assessed at every visit "through Week 24," which matches the objective of characterizing safety during treatment. The summary does not describe any safety follow-up after Week 24.

Measurement approach: Do the clinical assessments and participant-reported measures collect the type of data needed to evaluate the objectives? Explain.

Yes for efficacy. Disease activity is measured by clinical disease activity assessments (including joint examination) that produce the protocol-defined score; physical function by a validated participant questionnaire; and pain by a participant-reported 0–10 numeric rating scale. Each objective is therefore measured with the type of data it calls for: clinician-assessed for disease activity and participant-reported for function and pain.

Largely yes for safety. Adverse-event assessment, vital signs, safety laboratory testing, and concomitant medication review are collected at protocol-specified visits, which provides the data to characterize adverse events, serious adverse events, and laboratory abnormalities. The summary does not describe how injection-site reactions or treatment discontinuations are specifically captured.

Population fit: Does the target population match the population described in the research question and objectives? Explain.

Yes. The research question refers to "adults with moderate-to-severe rheumatoid arthritis who remain symptomatic despite stable standard therapy." Section 5 defines the target population in the same terms, and the inclusion criteria make each element operational: age 18 through 75, a documented diagnosis for at least 6 months, moderate-to-severe active disease at screening and baseline, and a stable dose of protocol-permitted background therapy for at least 8 weeks before randomization.

The exclusion criteria (active serious infection, tuberculosis, clinically significant immunodeficiency, pregnancy or breastfeeding, recent investigational product use, and significant laboratory

abnormalities or uncontrolled conditions) narrow the population for safety and interpretability without changing who the question is about. One point to note: the upper age limit of 75 means results will not describe older adults with rheumatoid arthritis.

Potential limitation or gap: Identify one methodology limitation, ambiguity, or area that would require clarification. If you believe the provided summary shows no clear gap, state that and explain why.

Section 7 lists a secondary endpoint, the "proportion of participants meeting the protocol-defined clinical response threshold at Week 24," that has no corresponding secondary objective in Section 3. The three stated secondary objectives cover physical function, pain, and safety and tolerability; none refers to clinical response.

The endpoint is clinically sensible and relates to the primary objective, since it expresses disease activity as a responder rate. As written, however, it is an endpoint without a stated objective, and the summary does not define the response threshold. This should be clarified in the full protocol.

Section 5: Overall Alignment Feedback

Overall assessment: Choose Strong Alignment, Partial Alignment, or Misalignment and provide a concise study-specific conclusion.

Strong Alignment for the efficacy objectives, with Partial Alignment for the safety and tolerability objective.

The primary objective (reducing disease activity from baseline to Week 24 compared with placebo) maps directly to the primary endpoint (change from baseline to Week 24 in the protocol-defined disease activity score), and the randomized, double-blind, placebo-controlled, parallel-group design supports a causal comparison. The physical-function and pain objectives map equally directly to the validated questionnaire score and the 0–10 numeric rating scale over the same interval.

Two gaps prevent a rating of strong alignment throughout. First, the safety and tolerability objective names injection-site reactions, clinically significant laboratory abnormalities, and treatment discontinuations, but the safety endpoint names only treatment-emergent adverse events and serious adverse events. Second, the clinical response threshold endpoint at Week 24 has no matching objective. Neither gap undermines the primary comparison, and the visit schedule and data collection approach are sufficient to support every stated objective.

Recommended clarification or improvement: Provide one specific recommendation based on your analysis. Do not invent a problem solely to fill this field; if the summary is sufficiently aligned, identify a detail that would be useful to confirm in the full protocol.

Before or during the site qualification visit, ask the sponsor to confirm that the full protocol and statistical analysis plan contain a one-to-one objective-to-endpoint map. Specifically: (1) whether separate safety endpoints exist for injection-site reactions, clinically significant laboratory abnormalities, and treatment discontinuations, or how these are captured within adverse-event reporting; and (2) which objective the "proportion of participants meeting the protocol-defined clinical response threshold at Week 24" supports and how the threshold is defined.

The answers determine what the site must document in source and enter in study systems at each visit, so they should be settled before the first participant is enrolled.

Final Self-Check

☑️ I used the Protocol Summary as the source for my analysis.

☑️ I mapped the primary objective to the correct primary endpoint.

☑️ I assessed at least two secondary objective-endpoint relationships.

☑️ I identified the data collection method and timing for the endpoints I analyzed.

☑️ My alignment ratings are supported by study-specific evidence.

☑️ My feedback explains why the elements align or identifies a specific gap.

☑️ I did not copy the model protocol's answers into the IMMU-201 analysis.

Training note: IMMU-201 and MIGRAINE-102 are synthetic studies created solely for this educational activity.